THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 3: **World of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of decimal places in the expansion of $\frac{3}{2^3 \times 5^2}$ is: 1

(a) 3 **(b)** 5 **(c)** 4 **(d)** 2

2. $x = 0.\overline{27}$ equals: 1

(a) $\frac{1}{4}$ **(b)** $\frac{2}{11}$ **(c)** $\frac{4}{11}$ **(d)** $\frac{3}{11}$

3. An irrational number between 2 and 3 is: 1

(a) $\sqrt{11}$ **(b)** $\sqrt{10}$ **(c)** $\sqrt{5}$ **(d)** $\sqrt{3}$

4. The sum of a rational number and an irrational number is: 1

(a) zero **(b)** rational **(c)** an integer **(d)** irrational

5. A rational number $\frac{p}{q}$ equal to $0.1\overline{6}$ is: 1

(a) $\frac{1}{3}$ **(b)** 0 **(c)** $\frac{1}{7}$ **(d)** $\frac{1}{6}$

6. $(-15) - (-23) =$ 1

(a) 8 **(b)** 7 **(c)** 10 **(d)** 9

7. Which of the following has a terminating decimal expansion? 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{7}$ **(c)** $\frac{5}{6}$ **(d)** $\frac{7}{16}$

8. The number of decimal places in the expansion of $\frac{7}{2^2 \times 5^4}$ is: 1

(a) 6 **(b)** 3 **(c)** 5 **(d)** 4

9. $x = 0.\overline{45}$ equals: 1

(a) $\frac{6}{11}$ **(b)** $\frac{5}{12}$ **(c)** $\frac{5}{11}$ **(d)** $\frac{4}{11}$

10. An irrational number between 3 and 4 is: 1

(a) $\sqrt{7}$ **(b)** $\sqrt{11}$ **(c)** $\sqrt{18}$ **(d)** $\sqrt{17}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Express $0.\overline{3}$ as a fraction in lowest terms. 2

12. Write $\frac{1}{6}$ as a decimal. 2

13. Find a rational number between 3 and 4. 2

14. Express $0.\overline{6}$ as a fraction in lowest terms. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Express $0.\overline{36}$ in the form $\frac{p}{q}$. **3**

16. Express $0.1\overline{6}$ in the form $\frac{p}{q}$. **3**

17. Find three rational numbers between $\frac{1}{3}$ and $\frac{2}{3}$. **3**

OR

Express $0.\overline{15}$ in the form $\frac{p}{q}$. **3**

18. Express $0.2\overline{7}$ in the form $\frac{p}{q}$. **3**

OR

Find three rational numbers between $\frac{1}{5}$ and $\frac{2}{5}$. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the decimal expansion of $\frac{1}{7}$ by long division and state the length of the repeating block. **5**

OR

Simplify $(5+\sqrt{3})(5-\sqrt{3})$ and $(5+\sqrt{3})^2$ and say which result is rational. **5**

20. Find the decimal expansion of $\frac{1}{13}$ by long division and state the length of the repeating block. **5**

OR

Simplify $(4+\sqrt{7})(4-\sqrt{7})$ and $(4+\sqrt{7})^2$ and say which result is rational. **5**

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND